Cinemática de una partícula en el espacio

Mecánica I · Tema 1 · Desarrollo simbólico

1. Coordenadas y base local

1. Producto vectorial.

$$\hat{\boldsymbol{e}}_j \times \hat{\boldsymbol{e}}_k = \varepsilon_{jki}\hat{\boldsymbol{e}}_i \implies \boldsymbol{b}\times\boldsymbol{c} = b_j c_k\,(\hat{\boldsymbol{e}}_j\times\hat{\boldsymbol{e}}_k) = \varepsilon_{ijk}\,b_j c_k\,\hat{\boldsymbol{e}}_i, \qquad \boxed{(\boldsymbol{b}\times\boldsymbol{c})_i = \varepsilon_{ijk}b_j c_k} \tag{1.1}$$

2. Leibniz. Lineal, alternada, $\det I = 1$:

$$\det(\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}) = a_i b_j c_k \det(\boldsymbol{e}_i,\boldsymbol{e}_j,\boldsymbol{e}_k), \qquad \det(\boldsymbol{e}_i,\boldsymbol{e}_j,\boldsymbol{e}_k) = \varepsilon_{ijk} \tag{1.2}$$

$$\boxed{\det(\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}) = \varepsilon_{ijk}a_i b_j c_k} \tag{1.3}$$

$$\varepsilon_{ijk}a_i b_j c_k = a_1b_2c_3 + a_2b_3c_1 + a_3b_1c_2 - a_1b_3c_2 - a_2b_1c_3 - a_3b_2c_1 \tag{1.4}$$

$$= a_1(b_2c_3 - b_3c_2) + a_2(b_3c_1 - b_1c_3) + a_3(b_1c_2 - b_2c_1) = a_i(\boldsymbol{b}\times\boldsymbol{c})_i \tag{1.5}$$

3. Producto mixto y volumen.

$$\boxed{\boldsymbol{a}\cdot(\boldsymbol{b}\times\boldsymbol{c}) = \varepsilon_{ijk}a_i b_j c_k = \det\begin{pmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{pmatrix}} \tag{1.6}$$

$$V = |\boldsymbol{b}\times\boldsymbol{c}|\cdot|\boldsymbol{a}\cdot\hat{\boldsymbol{n}}| = |\boldsymbol{b}\times\boldsymbol{c}|\,\frac{|\boldsymbol{a}\cdot(\boldsymbol{b}\times\boldsymbol{c})|}{|\boldsymbol{b}\times\boldsymbol{c}|} = |\boldsymbol{a}\cdot(\boldsymbol{b}\times\boldsymbol{c})| \tag{1.7}$$

4. Contracción ε–δ.

$$\boxed{\varepsilon_{kij}\,\varepsilon_{kmn} = \delta_{im}\delta_{jn} - \delta_{in}\delta_{jm}} \tag{1.8}$$

$$[\boldsymbol{a}\times(\boldsymbol{b}\times\boldsymbol{c})]_i = \varepsilon_{kij}\varepsilon_{kmn}a_j b_m c_n = (\delta_{im}\delta_{jn} - \delta_{in}\delta_{jm})a_j b_m c_n \tag{1.9}$$

$$\implies \boldsymbol{a}\times(\boldsymbol{b}\times\boldsymbol{c}) = \boldsymbol{b}(\boldsymbol{a}\cdot\boldsymbol{c}) - \boldsymbol{c}(\boldsymbol{a}\cdot\boldsymbol{b}) \tag{1.10}$$

5. Factores de escala. $\boldsymbol{r} = \boldsymbol{r}(q^1,q^2,q^3)$:

$$h_i = \left\|\frac{\partial\boldsymbol{r}}{\partial q^i}\right\|, \qquad \hat{\boldsymbol{e}}_i = \frac{1}{h_i}\frac{\partial\boldsymbol{r}}{\partial q^i}, \qquad \hat{\boldsymbol{e}}_i\cdot\hat{\boldsymbol{e}}_j = \delta_{ij}, \qquad \hat{\boldsymbol{e}}_1\times\hat{\boldsymbol{e}}_2 = \hat{\boldsymbol{e}}_3 \tag{1.11}$$

$$\mathrm{d}\boldsymbol{r} = \sum_i \frac{\partial\boldsymbol{r}}{\partial q^i}\,\mathrm{d}q^i = \sum_i h_i\,\mathrm{d}q^i\,\hat{\boldsymbol{e}}_i \tag{1.12}$$

$$\mathrm{d}s^2 = \Big(\sum_i h_i\,\mathrm{d}q^i\,\hat{\boldsymbol{e}}_i\Big)\cdot\Big(\sum_j h_j\,\mathrm{d}q^j\,\hat{\boldsymbol{e}}_j\Big) = \sum_{i,j} h_i h_j\,\mathrm{d}q^i\,\mathrm{d}q^j\,\underbrace{\hat{\boldsymbol{e}}_i\cdot\hat{\boldsymbol{e}}_j}_{\delta_{ij}} = \sum_i h_i^2(\mathrm{d}q^i)^2 \tag{1.13}$$

$$\boxed{\mathrm{d}s^2 = h_1^2(\mathrm{d}q^1)^2 + h_2^2(\mathrm{d}q^2)^2 + h_3^2(\mathrm{d}q^3)^2} \tag{1.14}$$

$$\text{no ortogonal: } \mathrm{d}s^2 = g_{ij}\,\mathrm{d}q^i\,\mathrm{d}q^j,\ g_{ij} = \partial_i\boldsymbol{r}\cdot\partial_j\boldsymbol{r} \tag{1.15}$$

6. Elementos diferenciales. (i,j,k) cíclica:

$$\mathrm{d}\boldsymbol{l}_i = h_i\,\mathrm{d}q^i\,\hat{\boldsymbol{e}}_i, \qquad \mathrm{d}\boldsymbol{S}_i = \mathrm{d}\boldsymbol{l}_j \times \mathrm{d}\boldsymbol{l}_k = h_j h_k\,\mathrm{d}q^j\,\mathrm{d}q^k\,\hat{\boldsymbol{e}}_i \tag{1.16}$$

$$\mathrm{d}V = (\mathrm{d}\boldsymbol{l}_1 \times \mathrm{d}\boldsymbol{l}_2)\cdot \mathrm{d}\boldsymbol{l}_3 = h_1 h_2 h_3\,\mathrm{d}q^1\,\mathrm{d}q^2\,\mathrm{d}q^3 \tag{1.17}$$

$$\partial_1\boldsymbol{r}\cdot(\partial_2\boldsymbol{r}\times\partial_3\boldsymbol{r}) = \begin{vmatrix} \partial_1 x & \partial_1 y & \partial_1 z \\ \partial_2 x & \partial_2 y & \partial_2 z \\ \partial_3 x & \partial_3 y & \partial_3 z \end{vmatrix} = \begin{vmatrix} \partial_1 x & \partial_2 x & \partial_3 x \\ \partial_1 y & \partial_2 y & \partial_3 y \\ \partial_1 z & \partial_2 z & \partial_3 z \end{vmatrix} = \frac{\partial(x,y,z)}{\partial(q^1,q^2,q^3)} \tag{1.18}$$

$$\boxed{J = \partial_1\boldsymbol{r}\cdot(\partial_2\boldsymbol{r}\times\partial_3\boldsymbol{r}) = h_1h_2h_3\,\hat{\boldsymbol{e}}_1\cdot(\hat{\boldsymbol{e}}_2\times\hat{\boldsymbol{e}}_3) = h_1h_2h_3} \tag{1.19}$$

$$J_{\text{cil}} = \begin{vmatrix} \cos\varphi & -\rho\sin\varphi & 0 \\ \sin\varphi & \rho\cos\varphi & 0 \\ 0 & 0 & 1 \end{vmatrix} = \rho, \qquad J_{\text{esf}} = \hat{\boldsymbol{e}}_r\cdot(r\,\hat{\boldsymbol{e}}_\theta \times r\sin\theta\,\hat{\boldsymbol{e}}_\varphi) = r^2\sin\theta \tag{1.20}$$

7. Cilíndricas. $x=\rho\cos\varphi$, $y=\rho\sin\varphi$, $z=z$:

$$\frac{\partial\boldsymbol{r}}{\partial\rho} = \cos\varphi\,\hat{\boldsymbol{\imath}} + \sin\varphi\,\hat{\boldsymbol{\jmath}}, \qquad \frac{\partial\boldsymbol{r}}{\partial\varphi} = \rho(-\sin\varphi\,\hat{\boldsymbol{\imath}} + \cos\varphi\,\hat{\boldsymbol{\jmath}}), \qquad \frac{\partial\boldsymbol{r}}{\partial z} = \hat{\boldsymbol{k}} \tag{1.21}$$

$$\boxed{h = (1,\rho,1), \quad \hat{\boldsymbol{e}}_\rho = \cos\varphi\,\hat{\boldsymbol{\imath}} + \sin\varphi\,\hat{\boldsymbol{\jmath}}, \quad \hat{\boldsymbol{e}}_\varphi = -\sin\varphi\,\hat{\boldsymbol{\imath}} + \cos\varphi\,\hat{\boldsymbol{\jmath}}} \tag{1.22}$$

$$\boldsymbol{r} = \rho\,\hat{\boldsymbol{e}}_\rho + z\,\hat{\boldsymbol{k}}, \qquad \mathrm{d}\boldsymbol{r} = \mathrm{d}\rho\,\hat{\boldsymbol{e}}_\rho + \rho\,\mathrm{d}\varphi\,\hat{\boldsymbol{e}}_\varphi + \mathrm{d}z\,\hat{\boldsymbol{k}}, \qquad \mathrm{d}V = \rho\,\mathrm{d}\rho\,\mathrm{d}\varphi\,\mathrm{d}z \tag{1.23}$$

$$\mathrm{d}\boldsymbol{S}_\rho = \rho\,\mathrm{d}\varphi\,\mathrm{d}z\,\hat{\boldsymbol{e}}_\rho, \qquad \mathrm{d}\boldsymbol{S}_\varphi = \mathrm{d}z\,\mathrm{d}\rho\,\hat{\boldsymbol{e}}_\varphi, \qquad \mathrm{d}\boldsymbol{S}_z = \rho\,\mathrm{d}\rho\,\mathrm{d}\varphi\,\hat{\boldsymbol{k}} \tag{1.24}$$

8. Esféricas. $x=r\sin\theta\cos\varphi$, $y=r\sin\theta\sin\varphi$, $z=r\cos\theta$:

$$\frac{\partial\boldsymbol{r}}{\partial r} = \hat{\boldsymbol{e}}_r, \qquad \frac{\partial\boldsymbol{r}}{\partial\theta} = r\,\hat{\boldsymbol{e}}_\theta, \qquad \frac{\partial\boldsymbol{r}}{\partial\varphi} = r\sin\theta\,\hat{\boldsymbol{e}}_\varphi \tag{1.25}$$

$$\begin{aligned} \hat{\boldsymbol{e}}_r &= \sin\theta\cos\varphi\,\hat{\boldsymbol{\imath}} + \sin\theta\sin\varphi\,\hat{\boldsymbol{\jmath}} + \cos\theta\,\hat{\boldsymbol{k}} \\ \hat{\boldsymbol{e}}_\theta &= \cos\theta\cos\varphi\,\hat{\boldsymbol{\imath}} + \cos\theta\sin\varphi\,\hat{\boldsymbol{\jmath}} - \sin\theta\,\hat{\boldsymbol{k}} \\ \hat{\boldsymbol{e}}_\varphi &= -\sin\varphi\,\hat{\boldsymbol{\imath}} + \cos\varphi\,\hat{\boldsymbol{\jmath}} \end{aligned} \qquad \boxed{h = (1,\,r,\,r\sin\theta)} \tag{1.26}$$

$$\boldsymbol{r} = r\,\hat{\boldsymbol{e}}_r, \qquad \mathrm{d}\boldsymbol{r} = \mathrm{d}r\,\hat{\boldsymbol{e}}_r + r\,\mathrm{d}\theta\,\hat{\boldsymbol{e}}_\theta + r\sin\theta\,\mathrm{d}\varphi\,\hat{\boldsymbol{e}}_\varphi, \qquad \mathrm{d}V = r^2\sin\theta\,\mathrm{d}r\,\mathrm{d}\theta\,\mathrm{d}\varphi \tag{1.27}$$

$$\mathrm{d}\boldsymbol{S}_r = r^2\sin\theta\,\mathrm{d}\theta\,\mathrm{d}\varphi\,\hat{\boldsymbol{e}}_r, \qquad \mathrm{d}\boldsymbol{S}_\theta = r\sin\theta\,\mathrm{d}\varphi\,\mathrm{d}r\,\hat{\boldsymbol{e}}_\theta, \qquad \mathrm{d}\boldsymbol{S}_\varphi = r\,\mathrm{d}r\,\mathrm{d}\theta\,\hat{\boldsymbol{e}}_\varphi \tag{1.28}$$

9. Derivadas de la base.

$$\mathrm{d}\hat{\boldsymbol{e}}_\rho = \mathrm{d}\varphi(-\sin\varphi\,\hat{\boldsymbol{\imath}} + \cos\varphi\,\hat{\boldsymbol{\jmath}}) = \mathrm{d}\varphi\,\hat{\boldsymbol{e}}_\varphi, \qquad \mathrm{d}\hat{\boldsymbol{e}}_\varphi = -\,\mathrm{d}\varphi(\cos\varphi\,\hat{\boldsymbol{\imath}} + \sin\varphi\,\hat{\boldsymbol{\jmath}}) = -\,\mathrm{d}\varphi\,\hat{\boldsymbol{e}}_\rho \tag{1.29}$$

$$\mathrm{d}\hat{\boldsymbol{e}}_r = \mathrm{d}\theta\,\hat{\boldsymbol{e}}_\theta + \sin\theta\,\mathrm{d}\varphi\,\hat{\boldsymbol{e}}_\varphi, \qquad \mathrm{d}\hat{\boldsymbol{e}}_\theta = -\,\mathrm{d}\theta\,\hat{\boldsymbol{e}}_r + \cos\theta\,\mathrm{d}\varphi\,\hat{\boldsymbol{e}}_\varphi \tag{1.30}$$

$$\mathrm{d}\hat{\boldsymbol{e}}_\varphi = -\,\mathrm{d}\varphi(\sin\theta\,\hat{\boldsymbol{e}}_r + \cos\theta\,\hat{\boldsymbol{e}}_\theta) \tag{1.31}$$

$$\boxed{\mathrm{d}\hat{\boldsymbol{e}}_i = \mathrm{d}\boldsymbol{\Omega}\times\hat{\boldsymbol{e}}_i, \qquad \mathrm{d}\boldsymbol{\Omega}_{\text{cil}} = \mathrm{d}\varphi\,\hat{\boldsymbol{k}}, \qquad \mathrm{d}\boldsymbol{\Omega}_{\text{esf}} = \mathrm{d}\varphi\,\hat{\boldsymbol{k}} + \mathrm{d}\theta\,\hat{\boldsymbol{e}}_\varphi} \tag{1.32}$$

$$\hat{\boldsymbol{k}} = \cos\theta\,\hat{\boldsymbol{e}}_r - \sin\theta\,\hat{\boldsymbol{e}}_\theta \tag{1.33}$$

10. Giro de la base: obtener $\mathrm{d}\boldsymbol{\Omega}$.

$$\mathrm{d}\hat{\boldsymbol{e}}_i = \sum_j A_{ij}\hat{\boldsymbol{e}}_j, \quad A_{ij} = \hat{\boldsymbol{e}}_j \cdot \mathrm{d}\hat{\boldsymbol{e}}_i \tag{1.34}$$

$$0 = \mathrm{d}(\hat{\boldsymbol{e}}_i \cdot \hat{\boldsymbol{e}}_j) = A_{ij} + A_{ji} \;\Rightarrow\; A_{ji} = -A_{ij},\; A_{ii} = 0 \tag{1.35}$$

$$\mathrm{d}\boldsymbol{\Omega} \times \hat{\boldsymbol{e}}_i = \sum_j M_{ij}\hat{\boldsymbol{e}}_j, \quad M_{ij} = \sum_k \varepsilon_{ijk}\Omega_k \tag{1.36}$$

$$M = A \iff \Omega_1 = A_{23},\; \Omega_2 = A_{31},\; \Omega_3 = A_{12} \tag{1.37}$$

$$\sum_{i,j} \varepsilon_{ijl}\varepsilon_{ijk} = 2\delta_{lk} \;\Rightarrow\; \Omega_l = \tfrac{1}{2}\sum_{i,j}\varepsilon_{ijl}A_{ij} \tag{1.38}$$

$$\sum_i \hat{\boldsymbol{e}}_i \times \mathrm{d}\hat{\boldsymbol{e}}_i = \sum_{i,j} A_{ij}\,\hat{\boldsymbol{e}}_i \times \hat{\boldsymbol{e}}_j = \sum_l \Big(\sum_{i,j}\varepsilon_{ijl}A_{ij}\Big)\hat{\boldsymbol{e}}_l = 2\,\mathrm{d}\boldsymbol{\Omega} \tag{1.39}$$

$$\boxed{\mathrm{d}\boldsymbol{\Omega} = \tfrac{1}{2}\sum_i \hat{\boldsymbol{e}}_i \times \mathrm{d}\hat{\boldsymbol{e}}_i} \tag{1.40}$$

cil.: $A_{\rho\varphi} = \mathrm{d}\varphi \;\Rightarrow\; \mathrm{d}\boldsymbol{\Omega} = \mathrm{d}\varphi\,\hat{\boldsymbol{k}}$; esf.: $2\,\mathrm{d}\boldsymbol{\Omega} = 2\,\mathrm{d}\theta\,\hat{\boldsymbol{e}}_\varphi + 2\,\mathrm{d}\varphi(\cos\theta\,\hat{\boldsymbol{e}}_r - \sin\theta\,\hat{\boldsymbol{e}}_\theta)$ (1.41)

2. Cinemática

1. Base móvil.

$$\boldsymbol{A} = \sum_i A_i\hat{\boldsymbol{e}}_i \;\Longrightarrow\; \frac{\mathrm{d}\boldsymbol{A}}{\mathrm{d}t} = \sum_i \dot{A}_i\,\hat{\boldsymbol{e}}_i + \sum_i A_i\,\dot{\hat{\boldsymbol{e}}}_i, \qquad \dot{\hat{\boldsymbol{e}}}_i = \boldsymbol{\Omega} \times \hat{\boldsymbol{e}}_i \tag{2.1}$$

2. Cilíndricas. $\dot{\hat{\boldsymbol{e}}}_\rho = \dot\varphi\,\hat{\boldsymbol{e}}_\varphi$, $\dot{\hat{\boldsymbol{e}}}_\varphi = -\dot\varphi\,\hat{\boldsymbol{e}}_\rho$:

$$\boldsymbol{v} = \frac{\mathrm{d}}{\mathrm{d}t}(\rho\,\hat{\boldsymbol{e}}_\rho + z\,\hat{\boldsymbol{k}}) = \dot\rho\,\hat{\boldsymbol{e}}_\rho + \rho\dot\varphi\,\hat{\boldsymbol{e}}_\varphi + \dot z\,\hat{\boldsymbol{k}} \tag{2.2}$$

$$\boldsymbol{a} = \ddot\rho\,\hat{\boldsymbol{e}}_\rho + \dot\rho\dot\varphi\,\hat{\boldsymbol{e}}_\varphi + (\dot\rho\dot\varphi + \rho\ddot\varphi)\hat{\boldsymbol{e}}_\varphi - \rho\dot\varphi^2\hat{\boldsymbol{e}}_\rho + \ddot z\,\hat{\boldsymbol{k}} \tag{2.3}$$

$$\boxed{\boldsymbol{a} = (\ddot\rho - \rho\dot\varphi^2)\,\hat{\boldsymbol{e}}_\rho + (\rho\ddot\varphi + 2\dot\rho\dot\varphi)\,\hat{\boldsymbol{e}}_\varphi + \ddot z\,\hat{\boldsymbol{k}}, \qquad a_\varphi = \frac{1}{\rho}\frac{\mathrm{d}}{\mathrm{d}t}(\rho^2\dot\varphi)} \tag{2.4}$$

3. Esféricas. $\dot{\hat{\boldsymbol{e}}}_r = \dot\theta\,\hat{\boldsymbol{e}}_\theta + \dot\varphi\sin\theta\,\hat{\boldsymbol{e}}_\varphi$, $\dot{\hat{\boldsymbol{e}}}_\theta = -\dot\theta\,\hat{\boldsymbol{e}}_r + \dot\varphi\cos\theta\,\hat{\boldsymbol{e}}_\varphi$, $\dot{\hat{\boldsymbol{e}}}_\varphi = -\dot\varphi(\sin\theta\,\hat{\boldsymbol{e}}_r + \cos\theta\,\hat{\boldsymbol{e}}_\theta)$:

$$\boldsymbol{v} = \dot r\,\hat{\boldsymbol{e}}_r + r\,\dot{\hat{\boldsymbol{e}}}_r = \dot r\,\hat{\boldsymbol{e}}_r + r\dot\theta\,\hat{\boldsymbol{e}}_\theta + r\sin\theta\,\dot\varphi\,\hat{\boldsymbol{e}}_\varphi \tag{2.5}$$

$$\boxed{\begin{aligned} a_r &= \ddot r - r\dot\theta^2 - r\sin^2\theta\,\dot\varphi^2 \\ a_\theta &= r\ddot\theta + 2\dot r\dot\theta - r\sin\theta\cos\theta\,\dot\varphi^2 = \tfrac{1}{r}\tfrac{\mathrm{d}}{\mathrm{d}t}(r^2\dot\theta) - r\sin\theta\cos\theta\,\dot\varphi^2 \\ a_\varphi &= r\sin\theta\,\ddot\varphi + 2\dot r\dot\varphi\sin\theta + 2r\dot\theta\dot\varphi\cos\theta = \tfrac{1}{r\sin\theta}\tfrac{\mathrm{d}}{\mathrm{d}t}(r^2\sin^2\theta\,\dot\varphi) \end{aligned}} \tag{2.6}$$

3. Triedro de Frenet

1. Longitud de arco.

$$s(t) = \int_{t_0}^{t} \|\dot{\boldsymbol{r}}\| \, \mathrm{d}t' = \int_{t_0}^{t} \sqrt{\dot{x}^2 + \dot{y}^2 + \dot{z}^2} \, \mathrm{d}t', \qquad \frac{\mathrm{d}s}{\mathrm{d}t} = v, \qquad \mathrm{d}s^2 = \mathrm{d}\boldsymbol{r} \cdot \mathrm{d}\boldsymbol{r} \tag{3.1}$$

2. Triedro.

$$\hat{\boldsymbol{\tau}} = \frac{\mathrm{d}\boldsymbol{r}}{\mathrm{d}s}, \qquad \kappa = \left\| \frac{\mathrm{d}\hat{\boldsymbol{\tau}}}{\mathrm{d}s} \right\|, \qquad \hat{\boldsymbol{n}} = \frac{1}{\kappa} \frac{\mathrm{d}\hat{\boldsymbol{\tau}}}{\mathrm{d}s}, \qquad \hat{\boldsymbol{b}} = \hat{\boldsymbol{\tau}} \times \hat{\boldsymbol{n}}, \qquad \rho_c = \frac{1}{\kappa} \tag{3.2}$$

3. Vector unitario.

$$\hat{\boldsymbol{a}} \cdot \hat{\boldsymbol{a}} = 1 \implies 0 = \frac{\mathrm{d}}{\mathrm{d}s}(\hat{\boldsymbol{a}} \cdot \hat{\boldsymbol{a}}) = 2\, \hat{\boldsymbol{a}} \cdot \frac{\mathrm{d}\hat{\boldsymbol{a}}}{\mathrm{d}s} \implies \hat{\boldsymbol{a}} \perp \hat{\boldsymbol{a}}' \tag{3.3}$$

4. Frenet–Serret.

$$\hat{\boldsymbol{b}}' = \hat{\boldsymbol{\tau}}' \times \hat{\boldsymbol{n}} + \hat{\boldsymbol{\tau}} \times \hat{\boldsymbol{n}}' = \kappa\, \hat{\boldsymbol{n}} \times \hat{\boldsymbol{n}} + \hat{\boldsymbol{\tau}} \times \hat{\boldsymbol{n}}' = \hat{\boldsymbol{\tau}} \times \hat{\boldsymbol{n}}' \perp \hat{\boldsymbol{\tau}},\ \hat{\boldsymbol{b}} \implies \hat{\boldsymbol{b}}' = -\mathcal{T}\, \hat{\boldsymbol{n}} \tag{3.4}$$

$$\hat{\boldsymbol{\tau}} \cdot \hat{\boldsymbol{n}}' = (\hat{\boldsymbol{\tau}} \cdot \hat{\boldsymbol{n}})' - \hat{\boldsymbol{\tau}}' \cdot \hat{\boldsymbol{n}} = -\kappa, \qquad \hat{\boldsymbol{n}} \cdot \hat{\boldsymbol{n}}' = 0, \qquad \hat{\boldsymbol{b}} \cdot \hat{\boldsymbol{n}}' = (\hat{\boldsymbol{b}} \cdot \hat{\boldsymbol{n}})' - \hat{\boldsymbol{b}}' \cdot \hat{\boldsymbol{n}} = \mathcal{T} \tag{3.5}$$

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}s} \begin{pmatrix} \hat{\boldsymbol{\tau}} \\ \hat{\boldsymbol{n}} \\ \hat{\boldsymbol{b}} \end{pmatrix} = \begin{pmatrix} 0 & \kappa & 0 \\ -\kappa & 0 & \mathcal{T} \\ 0 & -\mathcal{T} & 0 \end{pmatrix} \begin{pmatrix} \hat{\boldsymbol{\tau}} \\ \hat{\boldsymbol{n}} \\ \hat{\boldsymbol{b}} \end{pmatrix}, \qquad \frac{\mathrm{d}\hat{\boldsymbol{u}}}{\mathrm{d}s} = \boldsymbol{\omega} \times \hat{\boldsymbol{u}}, \quad \boldsymbol{\omega} = \mathcal{T}\, \hat{\boldsymbol{\tau}} + \kappa\, \hat{\boldsymbol{b}}} \tag{3.6}$$

5. Componentes intrínsecas.

$$\boldsymbol{v} = \frac{\mathrm{d}\boldsymbol{r}}{\mathrm{d}s} \frac{\mathrm{d}s}{\mathrm{d}t} = v\, \hat{\boldsymbol{\tau}}, \qquad \boldsymbol{a} = \dot{v}\, \hat{\boldsymbol{\tau}} + v \frac{\mathrm{d}\hat{\boldsymbol{\tau}}}{\mathrm{d}s} \frac{\mathrm{d}s}{\mathrm{d}t} \implies \boxed{\boldsymbol{a} = \dot{v}\, \hat{\boldsymbol{\tau}} + \kappa v^2\, \hat{\boldsymbol{n}}, \qquad a_T = \dot{v}, \quad a_N = \frac{v^2}{\rho_c}} \tag{3.7}$$

6. Parámetro arbitrario.

$$\boldsymbol{v} \times \boldsymbol{a} = v\, \hat{\boldsymbol{\tau}} \times (\dot{v}\, \hat{\boldsymbol{\tau}} + \kappa v^2\, \hat{\boldsymbol{n}}) = \kappa v^3\, \hat{\boldsymbol{b}}, \qquad \boldsymbol{v} \cdot \boldsymbol{a} = v\dot{v} \tag{3.8}$$

$$\dot{\boldsymbol{a}} = (\cdots)\hat{\boldsymbol{\tau}} + (\cdots)\hat{\boldsymbol{n}} + \kappa v^2\, \dot{\hat{\boldsymbol{n}}}, \qquad \dot{\hat{\boldsymbol{n}}} \cdot \hat{\boldsymbol{b}} = v\mathcal{T} \implies (\boldsymbol{v} \times \boldsymbol{a}) \cdot \dot{\boldsymbol{a}} = \kappa^2 v^6\, \mathcal{T} \tag{3.9}$$

$$\boxed{\begin{gathered} \hat{\boldsymbol{\tau}} = \frac{\boldsymbol{v}}{v}, \quad \hat{\boldsymbol{b}} = \frac{\boldsymbol{v} \times \boldsymbol{a}}{\|\boldsymbol{v} \times \boldsymbol{a}\|}, \quad \hat{\boldsymbol{n}} = \hat{\boldsymbol{b}} \times \hat{\boldsymbol{\tau}}, \\ \kappa = \frac{\|\boldsymbol{v} \times \boldsymbol{a}\|}{v^3}, \quad \mathcal{T} = \frac{(\boldsymbol{v} \times \boldsymbol{a}) \cdot \dot{\boldsymbol{a}}}{\|\boldsymbol{v} \times \boldsymbol{a}\|^2}, \quad a_T = \frac{\boldsymbol{v} \cdot \boldsymbol{a}}{v}, \quad a_N = \frac{\|\boldsymbol{v} \times \boldsymbol{a}\|}{v} \end{gathered}} \tag{3.10}$$

7. Circunferencia. $\theta = s/R$:

$$\boldsymbol{r} = R\big(\cos\tfrac{s}{R}, \sin\tfrac{s}{R}\big), \quad \hat{\boldsymbol{\tau}} = \big(-\sin\tfrac{s}{R}, \cos\tfrac{s}{R}\big), \quad \frac{\mathrm{d}\hat{\boldsymbol{\tau}}}{\mathrm{d}s} = -\frac{1}{R}\big(\cos\tfrac{s}{R}, \sin\tfrac{s}{R}\big) \tag{3.11}$$

$$\implies \kappa = \frac{1}{R}, \qquad a_N = \frac{v^2}{R} \tag{3.12}$$

4. Gradiente y Taylor

1. Diferencial y gradiente.

$$\mathrm{d}\Phi = \frac{\partial\Phi}{\partial x}\,\mathrm{d}x + \frac{\partial\Phi}{\partial y}\,\mathrm{d}y + \frac{\partial\Phi}{\partial z}\,\mathrm{d}z = \boldsymbol{\nabla}\Phi\cdot\mathrm{d}\boldsymbol{r}, \qquad \boldsymbol{\nabla}\Phi = \partial_x\Phi\,\hat{\boldsymbol{\imath}} + \partial_y\Phi\,\hat{\boldsymbol{\jmath}} + \partial_z\Phi\,\hat{\boldsymbol{k}} \tag{4.1}$$

2. Taylor.

$$f(\boldsymbol{q}) = \sum_{n=0}^{r}\frac{1}{n!}\Big[\Big(\sum_i \Delta q^i\frac{\partial}{\partial q^i}\Big)^n f\Big]_{\boldsymbol{q}_0} + O(\|\Delta\boldsymbol{q}\|^{r+1}) \tag{4.2}$$

$$\sin x\sin y = 0 + x\cdot 0 + y\cdot 0 + \tfrac{1}{2}\big[0 + 2xy\cos 0\cos 0 + 0\big] + O(3) = xy + O(3) \tag{4.3}$$

$$(x - \tfrac{x^3}{6} + \cdots)(y - \tfrac{y^3}{6} + \cdots) = xy - \tfrac{1}{6}(x^3y + xy^3) + \cdots \tag{4.4}$$

3. Derivada direccional. $\mathrm{d}\boldsymbol{r} = \hat{\boldsymbol{u}}\,\mathrm{d}s$:

$$\frac{\mathrm{d}\Phi}{\mathrm{d}s}\Big|_{\hat{\boldsymbol{u}}} = \boldsymbol{\nabla}\Phi\cdot\hat{\boldsymbol{u}} = \|\boldsymbol{\nabla}\Phi\|\cos\alpha \;\le\; \|\boldsymbol{\nabla}\Phi\|, \qquad \Phi = \text{const} \Rightarrow \boldsymbol{\nabla}\Phi\cdot\mathrm{d}\boldsymbol{r} = 0 \Rightarrow \boldsymbol{\nabla}\Phi \perp \{\Phi = c\} \tag{4.5}$$

4. Gradiente curvilíneo.

$$\mathrm{d}\Phi = \sum_i \frac{\partial\Phi}{\partial q^i}\,\mathrm{d}q^i = \boldsymbol{\nabla}\Phi\cdot\mathrm{d}\boldsymbol{r} = \sum_i G_i\,h_i\,\mathrm{d}q^i \implies G_i = \frac{1}{h_i}\frac{\partial\Phi}{\partial q^i} \tag{4.6}$$

$$\boxed{\boldsymbol{\nabla}\Phi = \sum_i \frac{1}{h_i}\frac{\partial\Phi}{\partial q^i}\hat{\boldsymbol{e}}_i} \tag{4.7}$$

$$\boldsymbol{\nabla}\Phi = \partial_\rho\Phi\,\hat{\boldsymbol{e}}_\rho + \tfrac{1}{\rho}\partial_\varphi\Phi\,\hat{\boldsymbol{e}}_\varphi + \partial_z\Phi\,\hat{\boldsymbol{k}} = \partial_r\Phi\,\hat{\boldsymbol{e}}_r + \tfrac{1}{r}\partial_\theta\Phi\,\hat{\boldsymbol{e}}_\theta + \tfrac{1}{r\sin\theta}\partial_\varphi\Phi\,\hat{\boldsymbol{e}}_\varphi \tag{4.8}$$

$$\boldsymbol{\nabla} f(r) = f'(r)\,\hat{\boldsymbol{e}}_r = f'(r)\,\frac{\boldsymbol{r}}{r} \tag{4.9}$$

5. Divergencia, rotacional y laplaciano

1. Continuidad.

$$\frac{\partial m}{\partial t} = \big[(\rho v_x)(x) - (\rho v_x)(x+\mathrm{d}x)\big]\,\mathrm{d}y\,\mathrm{d}z + \cdots = -\Big[\frac{\partial(\rho v_x)}{\partial x} + \frac{\partial(\rho v_y)}{\partial y} + \frac{\partial(\rho v_z)}{\partial z}\Big]\,\mathrm{d}V \tag{5.1}$$

$$m = \rho\,\mathrm{d}V \implies \boxed{\frac{\partial\rho}{\partial t} + \boldsymbol{\nabla}\cdot(\rho\boldsymbol{v}) = 0} \tag{5.2}$$

2. Divergencia curvilínea. $\boldsymbol{\nabla}\cdot\boldsymbol{A} = \lim \frac{1}{\Delta V}\oint \boldsymbol{A}\cdot \mathrm{d}\boldsymbol{S}$:

$$\mathrm{d}\Phi_{q^1} + \mathrm{d}\Phi_{q^1+\mathrm{d}q^1} = -[A_1h_2h_3]_{q^1}\,\mathrm{d}q^2\,\mathrm{d}q^3 + \left\{[A_1h_2h_3]_{q^1} + \frac{\partial}{\partial q^1}[A_1h_2h_3]\,\mathrm{d}q^1\right\}\mathrm{d}q^2\,\mathrm{d}q^3$$
$$= \frac{1}{h_1h_2h_3}\frac{\partial(A_1h_2h_3)}{\partial q^1}\,\mathrm{d}V \tag{5.3}$$

$$\boxed{\boldsymbol{\nabla}\cdot\boldsymbol{A} = \frac{1}{h_1h_2h_3}\left[\frac{\partial(A_1h_2h_3)}{\partial q^1} + \frac{\partial(A_2h_3h_1)}{\partial q^2} + \frac{\partial(A_3h_1h_2)}{\partial q^3}\right]} \tag{5.4}$$

$$\boldsymbol{\nabla}\cdot\boldsymbol{A} = \tfrac{1}{\rho}\partial_\rho(\rho A_\rho) + \tfrac{1}{\rho}\partial_\varphi A_\varphi + \partial_z A_z \tag{5.5}$$

$$\boldsymbol{\nabla}\cdot\boldsymbol{A} = \tfrac{1}{r^2}\partial_r(r^2A_r) + \tfrac{1}{r\sin\theta}\partial_\theta(\sin\theta A_\theta) + \tfrac{1}{r\sin\theta}\partial_\varphi A_\varphi \tag{5.6}$$

3. Rotacional cartesiano. Rectángulo $P(x,y)$, $R(x+\mathrm{d}x,y)$, $Q(x+\mathrm{d}x,y+\mathrm{d}y)$, $S(x,y+\mathrm{d}y)$:

$$C_{PRQ} = A_x\,\mathrm{d}x + \left[A_y + \partial_x A_y\,\mathrm{d}x\right]\mathrm{d}y, \qquad C_{PSQ} = A_y\,\mathrm{d}y + \left[A_x + \partial_y A_x\,\mathrm{d}y\right]\mathrm{d}x \tag{5.7}$$

$$C_{PRQ} - C_{PSQ} = (\partial_x A_y - \partial_y A_x)\,\mathrm{d}x\,\mathrm{d}y = (\boldsymbol{\nabla}\times\boldsymbol{A})\cdot(\mathrm{d}x\,\mathrm{d}y\,\hat{\boldsymbol{k}}) \tag{5.8}$$

$$\boxed{\boldsymbol{\nabla}\times\boldsymbol{A} = \begin{vmatrix}\hat{\boldsymbol{\imath}} & \hat{\boldsymbol{\jmath}} & \hat{\boldsymbol{k}}\\ \partial_x & \partial_y & \partial_z\\ A_x & A_y & A_z\end{vmatrix}, \qquad (\boldsymbol{\nabla}\times\boldsymbol{A})_i = \varepsilon_{ijk}\partial_j A_k} \tag{5.9}$$

4. Rotacional curvilíneo.

$$C_{PRQ} - C_{PSQ} = \left[\frac{\partial(A_2h_2)}{\partial q^1} - \frac{\partial(A_1h_1)}{\partial q^2}\right]\mathrm{d}q^1\,\mathrm{d}q^2 = \frac{1}{h_1h_2}\left[\frac{\partial(h_2A_2)}{\partial q^1} - \frac{\partial(h_1A_1)}{\partial q^2}\right]h_1h_2\,\mathrm{d}q^1\,\mathrm{d}q^2 \tag{5.10}$$

$$\boxed{\boldsymbol{\nabla}\times\boldsymbol{A} = \frac{1}{h_1h_2h_3}\begin{vmatrix}h_1\hat{\boldsymbol{e}}_1 & h_2\hat{\boldsymbol{e}}_2 & h_3\hat{\boldsymbol{e}}_3\\ \partial_1 & \partial_2 & \partial_3\\ h_1A_1 & h_2A_2 & h_3A_3\end{vmatrix}} \tag{5.11}$$

$$\boldsymbol{\nabla}\times\boldsymbol{A} = \frac{1}{\rho}\begin{vmatrix}\hat{\boldsymbol{e}}_\rho & \rho\,\hat{\boldsymbol{e}}_\varphi & \hat{\boldsymbol{k}}\\ \partial_\rho & \partial_\varphi & \partial_z\\ A_\rho & \rho A_\varphi & A_z\end{vmatrix}, \qquad \boldsymbol{\nabla}\times\boldsymbol{A} = \frac{1}{r^2\sin\theta}\begin{vmatrix}\hat{\boldsymbol{e}}_r & r\,\hat{\boldsymbol{e}}_\theta & r\sin\theta\,\hat{\boldsymbol{e}}_\varphi\\ \partial_r & \partial_\theta & \partial_\varphi\\ A_r & rA_\theta & r\sin\theta A_\varphi\end{vmatrix} \tag{5.12}$$

5. Rotación rígida. $\boldsymbol{v} = \boldsymbol{\omega}\times\boldsymbol{r}$, $v_i = \varepsilon_{ijk}\omega_j x_k$:

$$\boldsymbol{\nabla}\cdot\boldsymbol{v} = \varepsilon_{ijk}\omega_j\delta_{ik} = 0, \qquad (\boldsymbol{\nabla}\times\boldsymbol{v})_i = \varepsilon_{kij}\varepsilon_{kmn}\omega_m\delta_{jn} = (\delta_{im}\delta_{jj} - \delta_{ij}\delta_{jm})\omega_m = 3\omega_i - \omega_i \tag{5.13}$$

$$\boxed{\boldsymbol{\nabla}\times\boldsymbol{v} = 2\boldsymbol{\omega}} \tag{5.14}$$

6. Laplaciano. $A_i = h_i^{-1}\partial_i\Phi$ en la divergencia:

$$\boxed{\nabla^2\Phi = \frac{1}{h_1h_2h_3}\sum_{\text{cícl.}}\frac{\partial}{\partial q^1}\Big(\frac{h_2h_3}{h_1}\frac{\partial\Phi}{\partial q^1}\Big)} \tag{5.15}$$

$$\nabla^2\Phi = \tfrac{1}{\rho}\partial_\rho(\rho\,\partial_\rho\Phi) + \tfrac{1}{\rho^2}\partial_\varphi^2\Phi + \partial_z^2\Phi \tag{5.16}$$

$$\nabla^2\Phi = \tfrac{1}{r^2}\partial_r(r^2\partial_r\Phi) + \tfrac{1}{r^2\sin\theta}\partial_\theta(\sin\theta\,\partial_\theta\Phi) + \tfrac{1}{r^2\sin^2\theta}\partial_\varphi^2\Phi, \qquad \nabla^2 f(r) = f'' + \tfrac{2}{r}f' \tag{5.17}$$

7. Identidades nulas. Schwarz: $\partial_i\partial_j = \partial_j\partial_i$:

$$\boldsymbol{\nabla}\cdot(\boldsymbol{\nabla}\times\boldsymbol{A}) = \varepsilon_{ijk}\partial_i\partial_jA_k = 0, \qquad [\boldsymbol{\nabla}\times(\boldsymbol{\nabla}\Phi)]_i = \varepsilon_{ijk}\partial_j\partial_k\Phi = 0 \tag{5.18}$$

6. Integrales y teoremas

1. Integral de línea.

$$\int_C \boldsymbol{A}\cdot\mathrm{d}\boldsymbol{r} = \int_{t_0}^{t_1}\boldsymbol{A}(\boldsymbol{r}(t))\cdot\dot{\boldsymbol{r}}\;\mathrm{d}t = \int_P^Q\big(A_1h_1\,\mathrm{d}q^1 + A_2h_2\,\mathrm{d}q^2 + A_3h_3\,\mathrm{d}q^3\big) \tag{6.1}$$

2. Circulación. $\boldsymbol{A} = r\sin\theta\,\hat{\boldsymbol{e}}_r + 2r\,\hat{\boldsymbol{e}}_\varphi$, $r=1$, $\theta=\pi/2$:

$$\mathrm{d}\boldsymbol{r} = r\;\mathrm{d}\varphi\,\hat{\boldsymbol{e}}_\varphi \implies \boldsymbol{A}\cdot\mathrm{d}\boldsymbol{r} = 2r^2\,\mathrm{d}\varphi \implies \Gamma = \int_0^{2\pi}2r^2\,\mathrm{d}\varphi = 4\pi \tag{6.2}$$

3. Longitud. $x = A\cos\omega t$, $y = A\sin\omega t$, $z = vt$:

$$l = \int_0^{2\pi}\sqrt{\dot{x}^2+\dot{y}^2+\dot{z}^2}\;\mathrm{d}t = \int_0^{2\pi}\sqrt{(A\omega)^2+v^2}\;\mathrm{d}t = 2\pi\sqrt{(A\omega)^2+v^2} \tag{6.3}$$

4. Flujo.

$$\int_S\boldsymbol{A}\cdot\mathrm{d}\boldsymbol{S} = \iint_D\boldsymbol{A}\cdot\Big(\frac{\partial\boldsymbol{r}}{\partial u}\times\frac{\partial\boldsymbol{r}}{\partial v}\Big)\,\mathrm{d}u\,\mathrm{d}v, \qquad q^3=\text{const}:\;\Phi = \iint[A_3h_1h_2]\,\mathrm{d}q^1\,\mathrm{d}q^2 \tag{6.4}$$

5. Cuadrante.

$$S = \int_0^1\mathrm{d}y\int_0^{\sqrt{1-y^2}}\mathrm{d}x = \int_0^{\pi/2}\cos^2u\;\mathrm{d}u = \frac{\pi}{4}, \qquad S = \int_0^1\mathrm{d}\rho\int_0^{\pi/2}\rho\;\mathrm{d}\varphi = \frac{\pi}{4} \tag{6.5}$$

6. Semiesfera.

$$V = \int_0^R r^2\,\mathrm{d}r\int_0^{\pi/2}\sin\theta\;\mathrm{d}\theta\int_0^{2\pi}\mathrm{d}\varphi = \frac{R^3}{3}\cdot1\cdot2\pi = \frac{2\pi}{3}R^3 \tag{6.6}$$

$$V = 4\int_0^R\mathrm{d}x\int_0^a\sqrt{a^2-y^2}\;\mathrm{d}y$$

$$= 4\int_0^R\frac{\pi a^2}{4}\,\mathrm{d}x = \pi\int_0^R(R^2-x^2)\;\mathrm{d}x = \frac{2\pi}{3}R^3, \qquad a=\sqrt{R^2-x^2} \tag{6.7}$$

7. Gauss y Stokes.

$$\boxed{\oint_{\partial V} \boldsymbol{A}\cdot \mathrm{d}\boldsymbol{S} = \int_V \boldsymbol{\nabla}\cdot\boldsymbol{A}\,\mathrm{d}V, \qquad \oint_{\partial S} \boldsymbol{A}\cdot \mathrm{d}\boldsymbol{r} = \int_S (\boldsymbol{\nabla}\times\boldsymbol{A})\cdot \mathrm{d}\boldsymbol{S}} \tag{6.8}$$

8. Conservativos.

$$\boldsymbol{A} = \boldsymbol{\nabla}\Phi \implies \int_P^Q \boldsymbol{A}\cdot \mathrm{d}\boldsymbol{r} = \int_P^Q \mathrm{d}\Phi = \Phi(Q) - \Phi(P), \qquad \boldsymbol{\nabla}\times\boldsymbol{A} = \boldsymbol{\nabla}\times\boldsymbol{\nabla}\Phi = \boldsymbol{0} \tag{6.9}$$

$$\Phi(\boldsymbol{r}) = \int_{P_0}^{\boldsymbol{r}} \boldsymbol{A}\cdot \mathrm{d}\boldsymbol{r}' \implies \mathrm{d}\Phi = \boldsymbol{A}\cdot\mathrm{d}\boldsymbol{r} \implies \boldsymbol{A} = \boldsymbol{\nabla}\Phi \tag{6.10}$$

$$\boldsymbol{\nabla}\times\boldsymbol{A} = \boldsymbol{0},\ D \text{ simpl. conexo}: \oint_C \boldsymbol{A}\cdot\mathrm{d}\boldsymbol{r} = \int_S (\boldsymbol{\nabla}\times\boldsymbol{A})\cdot\mathrm{d}\boldsymbol{S} = 0 \tag{6.11}$$

$$\text{Contraejemplo: } \boldsymbol{A} = \frac{\hat{\boldsymbol{e}}_\varphi}{\rho}, \quad \boldsymbol{\nabla}\times\boldsymbol{A} = \frac{1}{\rho}\partial_\rho(\rho A_\varphi)\,\hat{\boldsymbol{k}} = \boldsymbol{0}, \quad \oint_{\rho=1} \boldsymbol{A}\cdot\mathrm{d}\boldsymbol{r} = \int_0^{2\pi} \frac{1}{\rho}\,\rho\,\mathrm{d}\varphi = 2\pi \tag{6.12}$$

9. Potencial por la poligonal.

$$\Phi(x,y,z) = \int_0^x A_x(s,0,0)\,\mathrm{d}s + \int_0^y A_y(x,s,0)\,\mathrm{d}s + \int_0^z A_z(x,y,s)\,\mathrm{d}s \tag{6.13}$$

7. Seminario

Problema 1. $\boldsymbol{v}$ y $\boldsymbol{a}$ en cartesianas, cilíndricas y esféricas

$$\boldsymbol{v} = \dot{x}\,\hat{\boldsymbol{\imath}} + \dot{y}\,\hat{\boldsymbol{\jmath}} + \dot{z}\,\hat{\boldsymbol{k}}, \qquad \boldsymbol{a} = \ddot{x}\,\hat{\boldsymbol{\imath}} + \ddot{y}\,\hat{\boldsymbol{\jmath}} + \ddot{z}\,\hat{\boldsymbol{k}} \tag{7.1}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}(\dot\rho\,\hat{\boldsymbol{e}}_\rho) = \ddot\rho\,\hat{\boldsymbol{e}}_\rho + \dot\rho\dot\varphi\,\hat{\boldsymbol{e}}_\varphi, \quad \frac{\mathrm{d}}{\mathrm{d}t}(\rho\dot\varphi\,\hat{\boldsymbol{e}}_\varphi) = (\dot\rho\dot\varphi + \rho\ddot\varphi)\hat{\boldsymbol{e}}_\varphi - \rho\dot\varphi^2\hat{\boldsymbol{e}}_\rho, \quad \frac{\mathrm{d}}{\mathrm{d}t}(\dot z\,\hat{\boldsymbol{k}}) = \ddot z\,\hat{\boldsymbol{k}} \tag{7.2}$$

$$\boxed{\boldsymbol{v} = \dot\rho\,\hat{\boldsymbol{e}}_\rho + \rho\dot\varphi\,\hat{\boldsymbol{e}}_\varphi + \dot z\,\hat{\boldsymbol{k}}, \qquad \boldsymbol{a} = (\ddot\rho - \rho\dot\varphi^2)\hat{\boldsymbol{e}}_\rho + (\rho\ddot\varphi + 2\dot\rho\dot\varphi)\hat{\boldsymbol{e}}_\varphi + \ddot z\,\hat{\boldsymbol{k}}} \tag{7.3}$$

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}(\dot r\,\hat{\boldsymbol{e}}_r) &= \ddot r\,\hat{\boldsymbol{e}}_r + \dot r\dot\theta\,\hat{\boldsymbol{e}}_\theta + \dot r\dot\varphi\sin\theta\,\hat{\boldsymbol{e}}_\varphi \\
\frac{\mathrm{d}}{\mathrm{d}t}(r\dot\theta\,\hat{\boldsymbol{e}}_\theta) &= (\dot r\dot\theta + r\ddot\theta)\hat{\boldsymbol{e}}_\theta - r\dot\theta^2\hat{\boldsymbol{e}}_r + r\dot\theta\dot\varphi\cos\theta\,\hat{\boldsymbol{e}}_\varphi \\
\frac{\mathrm{d}}{\mathrm{d}t}(r\sin\theta\,\dot\varphi\,\hat{\boldsymbol{e}}_\varphi) &= (\dot r\sin\theta\,\dot\varphi + r\cos\theta\,\dot\theta\dot\varphi + r\sin\theta\,\ddot\varphi)\hat{\boldsymbol{e}}_\varphi - r\sin\theta\,\dot\varphi^2(\sin\theta\,\hat{\boldsymbol{e}}_r + \cos\theta\,\hat{\boldsymbol{e}}_\theta)
\end{aligned} \tag{7.4}$$

$$\boxed{\begin{aligned}
\boldsymbol{v} &= \dot r\,\hat{\boldsymbol{e}}_r + r\dot\theta\,\hat{\boldsymbol{e}}_\theta + r\sin\theta\,\dot\varphi\,\hat{\boldsymbol{e}}_\varphi \\
\boldsymbol{a} &= (\ddot r - r\dot\theta^2 - r\sin^2\theta\,\dot\varphi^2)\hat{\boldsymbol{e}}_r + (r\ddot\theta + 2\dot r\dot\theta - r\sin\theta\cos\theta\,\dot\varphi^2)\hat{\boldsymbol{e}}_\theta \\
&\quad + (r\sin\theta\,\ddot\varphi + 2\dot r\dot\varphi\sin\theta + 2r\dot\theta\dot\varphi\cos\theta)\hat{\boldsymbol{e}}_\varphi
\end{aligned}} \tag{7.5}$$

Problema 2. Funciones radiales

$$r^2 = x^2 + y^2 + z^2 \Rightarrow 2r\,\partial_x r = 2x \Rightarrow \boldsymbol{\nabla} r = \frac{\boldsymbol{r}}{r} \tag{7.6}$$

$$\boxed{\boldsymbol{\nabla} f(r) = f'(r)\boldsymbol{\nabla} r = \frac{\boldsymbol{r}}{r}\frac{\mathrm{d}f}{\mathrm{d}r}}, \qquad \boxed{\boldsymbol{\nabla} r^n = n\,r^{n-1}\frac{\boldsymbol{r}}{r} = n\,r^{n-2}\,\boldsymbol{r}} \tag{7.7}$$

$$\nabla^2 \ln r = \frac{1}{r^2}\frac{\mathrm{d}}{\mathrm{d}r}\left(r^2 \cdot \frac{1}{r}\right) = \frac{1}{r^2} \tag{7.8}$$

$$\boldsymbol{\nabla} \cdot \left(\frac{\boldsymbol{r}}{r^2}\right) = \frac{\boldsymbol{\nabla}\cdot\boldsymbol{r}}{r^2} + \boldsymbol{r}\cdot\left(-2r^{-4}\boldsymbol{r}\right) = \frac{3}{r^2} - \frac{2}{r^2} = \boxed{\frac{1}{r^2}} \tag{7.9}$$

Problema 3. Tobogán $\boldsymbol{r} = 2t\,\hat{\boldsymbol{\imath}} + (3 - \frac{1}{2}t^2)\hat{\boldsymbol{k}}$

$$\boxed{\boldsymbol{v} = 2\,\hat{\boldsymbol{\imath}} - t\,\hat{\boldsymbol{k}}, \qquad \boldsymbol{a} = -\hat{\boldsymbol{k}}}, \qquad v = \sqrt{4 + t^2} \tag{7.10}$$

$$t = 1:\ \hat{\boldsymbol{T}} = \frac{2\,\hat{\boldsymbol{\imath}} - \hat{\boldsymbol{k}}}{\sqrt{5}}, \qquad a_T = \frac{\boldsymbol{v}\cdot\boldsymbol{a}}{v} = \frac{t}{\sqrt{4+t^2}} = \frac{1}{\sqrt{5}} \tag{7.11}$$

$$a_N\hat{\boldsymbol{N}} = \boldsymbol{a} - a_T\hat{\boldsymbol{T}} = -\hat{\boldsymbol{k}} - \frac{2\,\hat{\boldsymbol{\imath}} - \hat{\boldsymbol{k}}}{5} = -\frac{2}{5}\,\hat{\boldsymbol{\imath}} - \frac{4}{5}\,\hat{\boldsymbol{k}}$$

$$\Longrightarrow\ a_N = \frac{2}{\sqrt{5}}, \qquad \boxed{\hat{\boldsymbol{N}} = -\frac{\hat{\boldsymbol{\imath}} + 2\,\hat{\boldsymbol{k}}}{\sqrt{5}}}, \qquad \hat{\boldsymbol{B}} = \hat{\boldsymbol{\jmath}} \tag{7.12}$$

$$\boldsymbol{v}\times\boldsymbol{a} = \begin{vmatrix} \hat{\boldsymbol{\imath}} & \hat{\boldsymbol{\jmath}} & \hat{\boldsymbol{k}} \\ 2 & 0 & -t \\ 0 & 0 & -1 \end{vmatrix} = 2\,\hat{\boldsymbol{\jmath}}$$

$$\boxed{\kappa = \frac{2}{(4+t^2)^{3/2}}, \quad \kappa(1) = \frac{2}{5\sqrt{5}}, \quad R_c(1) = \frac{5\sqrt{5}}{2} \simeq 5{,}59} \tag{7.13}$$

$$\boxed{a_T = \frac{t}{\sqrt{4+t^2}}, \qquad a_N = \kappa v^2 = \frac{2}{\sqrt{4+t^2}}}, \qquad a_T^2 + a_N^2 = 1 = \|\boldsymbol{a}\|^2 \tag{7.14}$$

Problema 4. Cilíndricas parabólicas $x = \frac{1}{2}(u^2 - v^2)$, $y = uv$, $z = z$

$$\frac{\partial \boldsymbol{r}}{\partial u} = u\,\hat{\boldsymbol{\imath}} + v\,\hat{\boldsymbol{\jmath}}, \qquad \frac{\partial \boldsymbol{r}}{\partial v} = -v\,\hat{\boldsymbol{\imath}} + u\,\hat{\boldsymbol{\jmath}}, \qquad \frac{\partial \boldsymbol{r}}{\partial z} = \hat{\boldsymbol{k}}, \qquad \partial_u \boldsymbol{r} \cdot \partial_v \boldsymbol{r} = -uv + vu = 0 \tag{7.15}$$

$$\boxed{h_u = h_v = \sqrt{u^2 + v^2}, \quad h_z = 1}, \qquad \hat{\boldsymbol{e}}_u = \frac{u\,\hat{\boldsymbol{\imath}} + v\,\hat{\boldsymbol{\jmath}}}{\sqrt{u^2 + v^2}}, \quad \hat{\boldsymbol{e}}_v = \frac{-v\,\hat{\boldsymbol{\imath}} + u\,\hat{\boldsymbol{\jmath}}}{\sqrt{u^2 + v^2}} \tag{7.16}$$

$$\boxed{\boldsymbol{\nabla} = \frac{1}{\sqrt{u^2 + v^2}}\left(\hat{\boldsymbol{e}}_u\,\partial_u + \hat{\boldsymbol{e}}_v\,\partial_v\right) + \hat{\boldsymbol{k}}\,\partial_z} \tag{7.17}$$

$$\mathrm{d}\boldsymbol{l}_u = \sqrt{u^2 + v^2}\,\mathrm{d}u\,\hat{\boldsymbol{e}}_u, \quad \mathrm{d}\boldsymbol{l}_v = \sqrt{u^2 + v^2}\,\mathrm{d}v\,\hat{\boldsymbol{e}}_v, \quad \mathrm{d}\boldsymbol{l}_z = \mathrm{d}z\,\hat{\boldsymbol{k}} \tag{7.18}$$

$$\mathrm{d}\boldsymbol{S}_u = \sqrt{u^2 + v^2}\,\mathrm{d}v\,\mathrm{d}z\,\hat{\boldsymbol{e}}_u, \quad \mathrm{d}\boldsymbol{S}_v = \sqrt{u^2 + v^2}\,\mathrm{d}z\,\mathrm{d}u\,\hat{\boldsymbol{e}}_v, \quad \mathrm{d}\boldsymbol{S}_z = (u^2 + v^2)\,\mathrm{d}u\,\mathrm{d}v\,\hat{\boldsymbol{k}} \tag{7.19}$$

$$\boxed{\mathrm{d}V = (u^2 + v^2)\,\mathrm{d}u\,\mathrm{d}v\,\mathrm{d}z}, \qquad \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} u & -v \\ v & u \end{vmatrix} = u^2 + v^2 \tag{7.20}$$

Problema 5. Potencial de $\boldsymbol{A} = 2xy\,\hat{\boldsymbol{\imath}} + (x^2 + 2yz^3)\hat{\boldsymbol{\jmath}} + 3y^2z^2\hat{\boldsymbol{k}}$

$$\boldsymbol{\nabla} \times \boldsymbol{A} = (6yz^2 - 6yz^2)\hat{\boldsymbol{\imath}} + (0 - 0)\hat{\boldsymbol{\jmath}} + (2x - 2x)\hat{\boldsymbol{k}} = \boldsymbol{0} \tag{7.21}$$

$$\partial_x \Phi = 2xy \Rightarrow \Phi = x^2 y + g(y, z), \qquad x^2 + \partial_y g = x^2 + 2yz^3 \Rightarrow g = y^2 z^3 + h(z), \qquad h'(z) = 0 \tag{7.22}$$

$$\boxed{\Phi = x^2 y + y^2 z^3 + C}, \qquad \int_0^x 0\,\mathrm{d}s + \int_0^y x^2\,\mathrm{d}s + \int_0^z 3y^2 s^2\,\mathrm{d}s = x^2 y + y^2 z^3 \tag{7.23}$$

Problema 6. Identidades nulas

$$\boldsymbol{\nabla} \cdot (\boldsymbol{\nabla} \times \boldsymbol{A}) = \partial_i(\varepsilon_{ijk}\partial_j A_k) = \varepsilon_{ijk}\partial_i\partial_j A_k = S, \qquad S \overset{i \leftrightarrow j}{=} \varepsilon_{jik}\partial_j\partial_i A_k = -S \;\Rightarrow\; \boxed{S = 0} \tag{7.24}$$

$$\partial_x(\partial_y A_z - \partial_z A_y) + \partial_y(\partial_z A_x - \partial_x A_z) + \partial_z(\partial_x A_y - \partial_y A_x) = 0 \tag{7.25}$$

$$[\boldsymbol{\nabla} \times (\boldsymbol{\nabla}\Phi)]_i = \varepsilon_{ijk}\partial_j\partial_k\Phi = 0, \qquad [\boldsymbol{\nabla} \times (\boldsymbol{\nabla}\Phi)]_x = \partial_y\partial_z\Phi - \partial_z\partial_y\Phi = 0 \tag{7.26}$$

Problema 7. Hélice $\boldsymbol{r} = A\cos\theta\,\hat{\boldsymbol{\imath}} + A\sin\theta\,\hat{\boldsymbol{\jmath}} + \frac{b}{2\pi}\theta\,\hat{\boldsymbol{k}}$, $c = b/2\pi$

$$\boldsymbol{v} = \dot{\theta}\,(-A\sin\theta,\, A\cos\theta,\, c), \qquad v = \dot{\theta}\sqrt{A^2 + c^2} \tag{7.27}$$

$$\boxed{\boldsymbol{a} = \ddot{\theta}\,(-A\sin\theta\,\hat{\boldsymbol{\imath}} + A\cos\theta\,\hat{\boldsymbol{\jmath}} + c\,\hat{\boldsymbol{k}}) - A\dot{\theta}^2(\cos\theta\,\hat{\boldsymbol{\imath}} + \sin\theta\,\hat{\boldsymbol{\jmath}})} \tag{7.28}$$

$$a_T = \ddot{\theta}\sqrt{A^2 + c^2}, \qquad a_N = A\dot{\theta}^2 = \kappa v^2 \;\Rightarrow\; \kappa = \frac{A}{A^2 + c^2}, \qquad \hat{\boldsymbol{n}} = -\hat{\boldsymbol{e}}_\rho \tag{7.29}$$

$$\boxed{s(\theta) = \int_0^\theta \left\| \frac{\mathrm{d}\boldsymbol{r}}{\mathrm{d}\theta'} \right\| \mathrm{d}\theta' = \theta\sqrt{A^2 + \frac{b^2}{4\pi^2}}, \qquad s_{\text{vuelta}} = \sqrt{(2\pi A)^2 + b^2}} \tag{7.30}$$

Problema 8. Espiral cónica $\boldsymbol{r} = \mathrm{e}^t(\cos t, \sin t, 1)$, $0 \le t \le 2$

$$\dot{x} = \mathrm{e}^t(\cos t - \sin t), \quad \dot{y} = \mathrm{e}^t(\sin t + \cos t), \quad \dot{z} = \mathrm{e}^t \tag{7.31}$$

$$\dot{x}^2 + \dot{y}^2 = \mathrm{e}^{2t}\big[(\cos t - \sin t)^2 + (\sin t + \cos t)^2\big] = 2\mathrm{e}^{2t} \;\Rightarrow\; \|\dot{\boldsymbol{r}}\| = \sqrt{3}\,\mathrm{e}^t \tag{7.32}$$

$$\boxed{s = \int_0^2 \sqrt{3}\,\mathrm{e}^t \,\mathrm{d}t = \sqrt{3}\,(\mathrm{e}^2 - 1) \simeq 11{,}07} \tag{7.33}$$

$$\rho = z = \mathrm{e}^t,\ \varphi = t:\ \dot{s}^2 = \dot{\rho}^2 + \rho^2\dot{\varphi}^2 + \dot{z}^2 = 3\mathrm{e}^{2t} \tag{7.34}$$

Problema 9. Triedro de Frenet: circunferencia, elipse, hélice

$$\boldsymbol{r} = R(\cos t, \sin t, 0), \qquad \boldsymbol{v} \times \boldsymbol{a} = R^2\hat{\boldsymbol{k}} \tag{7.35}$$

$$\boxed{\hat{\boldsymbol{\tau}} = (-\sin t, \cos t, 0),\ \hat{\boldsymbol{n}} = -(\cos t, \sin t, 0),\ \hat{\boldsymbol{b}} = \hat{\boldsymbol{k}},\ \kappa = \tfrac{1}{R},\ \mathcal{T} = 0} \tag{7.36}$$

$$\boldsymbol{r} = (a\cos t, b\sin t, 0), \qquad \boldsymbol{v} = (-a\sin t, b\cos t, 0) \tag{7.37}$$

$$\boldsymbol{a} = (-a\cos t, -b\sin t, 0), \qquad \boldsymbol{v} \times \boldsymbol{a} = ab\,\hat{\boldsymbol{k}} \tag{7.38}$$

$$\sigma = \sqrt{a^2\sin^2 t + b^2\cos^2 t} \tag{7.39}$$

$$\boxed{\hat{\boldsymbol{\tau}} = \frac{(-a\sin t,\ b\cos t,\ 0)}{\sigma},\ \hat{\boldsymbol{n}} = -\frac{(b\cos t,\ a\sin t,\ 0)}{\sigma},\ \hat{\boldsymbol{b}} = \hat{\boldsymbol{k}},\ \kappa = \frac{ab}{\sigma^3},\ \mathcal{T} = 0} \tag{7.40}$$

$$\boldsymbol{r} = (R\cos t, R\sin t, ct), \qquad c = \tfrac{b}{2\pi}, \qquad L = \sqrt{R^2 + c^2} \tag{7.41}$$

$$\boldsymbol{v} \times \boldsymbol{a} = \begin{vmatrix} \hat{\boldsymbol{\imath}} & \hat{\boldsymbol{\jmath}} & \hat{\boldsymbol{k}} \\ -R\sin t & R\cos t & c \\ -R\cos t & -R\sin t & 0 \end{vmatrix} = (Rc\sin t, -Rc\cos t, R^2) \tag{7.42}$$

$$\|\boldsymbol{v} \times \boldsymbol{a}\| = RL, \qquad (\boldsymbol{v} \times \boldsymbol{a}) \cdot \dot{\boldsymbol{a}} = (\boldsymbol{v} \times \boldsymbol{a}) \cdot (R\sin t, -R\cos t, 0) = R^2 c \tag{7.43}$$

$$\boxed{\begin{gathered} \hat{\boldsymbol{\tau}} = \frac{(-R\sin t,\ R\cos t,\ c)}{L}, \quad \hat{\boldsymbol{n}} = -(\cos t,\ \sin t,\ 0), \\ \hat{\boldsymbol{b}} = \frac{(c\sin t,\ -c\cos t,\ R)}{L}, \\ \kappa = \frac{R}{R^2 + c^2} = \frac{4\pi^2 R}{4\pi^2 R^2 + b^2}, \qquad \mathcal{T} = \frac{c}{R^2 + c^2} = \frac{2\pi b}{4\pi^2 R^2 + b^2} \end{gathered}} \tag{7.44}$$

Problema 10. Gradientes

$$\boxed{\boldsymbol{\nabla}(x^2 + y^2 + z^2) = 2\boldsymbol{r}}, \qquad \boxed{\boldsymbol{\nabla}\frac{1}{r} = -\frac{\hat{\boldsymbol{e}}_r}{r^2} = -\frac{\boldsymbol{r}}{r^3}} \tag{7.45}$$

$$\boxed{\boldsymbol{\nabla}(x\cos x + xyz) = (\cos x - x\sin x + yz)\,\hat{\boldsymbol{\imath}} + xz\,\hat{\boldsymbol{\jmath}} + xy\,\hat{\boldsymbol{k}}} \tag{7.46}$$

Problema 11. Factores de escala

$$\mathrm{d}x^2 + \mathrm{d}y^2 = (\cos\varphi\ \mathrm{d}\rho - \rho\sin\varphi\ \mathrm{d}\varphi)^2 + (\sin\varphi\ \mathrm{d}\rho + \rho\cos\varphi\ \mathrm{d}\varphi)^2 = \mathrm{d}\rho^2 + \rho^2\,\mathrm{d}\varphi^2 \tag{7.47}$$

$$\|\partial_r \boldsymbol{r}\| = 1, \qquad \|\partial_\theta \boldsymbol{r}\| = r, \qquad \|\partial_\varphi \boldsymbol{r}\| = r\sin\theta \tag{7.48}$$

$$\boxed{h_{\text{cart}} = (1,1,1), \qquad h_{\text{cil}} = (1,\rho,1), \qquad h_{\text{esf}} = (1, r, r\sin\theta)} \tag{7.49}$$